第九周作业

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1. **习题四:**

28.

设 $P(A) = p_A, P(B) = p_B$

则 $E(X) = 1 \times P(A) + 0 \times P(\overline{A}) = p_A, E(Y) = 1 \times P(B) + 0 \times P(\overline{B}) = p_B$

$E(XY) = 1 \times P(AB) + 0 \times (1 - P(AB)) = P(AB)$

$\therefore Cov(X, Y) = E(XY) - E(X)E(Y) = P(AB) - p_A p_B$

若 X 与 Y 不相关，则 $Cov(X, Y) = 0, P(AB) = p_A p_B$

$\therefore A, B$ 相互独立

$\therefore P(X = 1, Y = 1) = P(AB) = p_A p_B = p(X = 1)P(Y = 1), P(X = 1, Y = 0) = P(A\overline{B}) = p_A(1 - p_B) = P(X = 1)P(Y = 0), P(X = 0, Y = 1) = P(B\overline{A}) = p_B(1 - p_A) = P(X = 0)P(Y = 1), P(X = 0, Y = 0) = P(\overline{AB}) = (1 - p_A)(1 - p_B) = P(X = 0)P(Y = 0)$

$\therefore X, Y$ 相互独立

29.

由题意得 $X \sim N(1, 9), Y \sim N(0, 16)$

$E(Z) = E(\frac{X}{3} + \frac{Y}{2}) = \frac{E(X)}{3} + \frac{E(Y)}{2} = \frac{1}{3} + \frac{0}{2} = \frac{1}{3}$

$Cov(X, Y) = \rho_{X,Y} \times \sqrt{D(X)} \times \sqrt{D(Y)} = -6$

$D(Z) = \frac{1}{9}D(X) + \frac{2}{6}Cov(X, Y) + \frac{1}{4}D(Y) = 3$

$Cov(X, Z) = Cov(X, \frac{X}{3} + \frac{Y}{2}) = \frac{1}{3}D(X) + \frac{1}{2}Cov(X, Y) = 0$

$\therefore \rho_{X,Z} = \frac{Cov(X,Z)}{\sqrt{D(X)}\sqrt{D(Z)}} = 0$

31.

由题意得 $D(X) = 1, Cov(X, Y) = 2, D(Y) = 5$

$D(U) = D(X) - 4Cov(X,Y) + 4D(Y) = 13, D(V) = 4D(X) - 4Cov(X,Y) + D(Y) = 1$

$Cov(U,V) = 2D(X) - 5Cov(X,Y) + 2D(Y) = 2$

$\therefore \rho_{UV} = \frac{Cov(U,V)}{\sqrt{D(U)}\sqrt{D(V)}} = \frac{2\sqrt{13}}{13}$

2. **4.3-6 补充题一:**

由题意得

$(1) X \sim D(0,1), Y \sim D(0,1)$

$Cov(X,Y) = \rho_{X,Y} \times D(X) \times D(Y) = \rho$

$Cov(X, X+aY) = D(X) + aCov(X,Y) = 1 + a\rho = 0$

$\therefore a = -\frac{1}{\rho}$

(2) 令 $Z = X + aY$

$E(Z) = E(X) + aE(Y) = 0, D(Z) = D(X) + 2aCov(X,Y) + a^2 D(Y) = \frac{1}{\rho^2}$

$\therefore Z \sim N(0, \frac{1}{\rho^2}), f(z) = \frac{\rho}{\sqrt{2\pi}} e^{-\frac{x^2\rho^2}{2}}$

$\therefore E(|Z|) = \int_{-\infty}^{+\infty} |z| f_Z(z) dz = \frac{2\rho}{\sqrt{2\pi}} \int_0^{+\infty} z e^{-\frac{z^2\rho^2}{2}} dx = \frac{\sqrt{2}\rho}{\sqrt{\pi}}$

$\therefore E(|X + aY|) = \frac{\sqrt{2}\rho}{\sqrt{\pi}}$

4.3-6 补充题二:

$\because X_1, ..., X_n$ 相互独立，则 $Cov(X_i, X_j) = 0, i, j = 1, 2, ..., n$

因为 X_i 服从泊松分布, 故 $E(X_i) = \lambda, D(X_i) = \lambda$

$Cov(Y,Z) = \sum_{i=2}^{n-1} D(X_i) = \lambda(n-2)$

3. **习题五:**

2.

设 X = ” 每毫升血液中所含的白细胞数”，$E(X) = 7300, D(X) = 490000$

$P(5200 \leq X \leq 9400) = P(|X - E(X)| \leq 2100) \geq 1 - \frac{D(X)}{2100^2} = \frac{8}{9}$

5.

因为 $\{X_n\}$ 相互独立

令 $Y_n = \frac{1}{n}\sum_{k=1}^{n} X_k, E(Y_n) = \frac{1}{n}\sum_{k=1}^{n} E(X_n) = \frac{1}{n}\sum_{k=1}^{n}(-\sqrt{lnn} \times 0.5 + \sqrt{lnn} \times 0.5) = 0$

$D(X_n) = (-\sqrt{lnn} - 0)^2 \times 0.5 + (\sqrt{lnn} - 0) \times 0.5 = lnn\ D(Y_n) = D(\frac{1}{n}\sum_{k=1}^{n} X_k) = \frac{1}{n^2}\sum_{k=1}^{n} D(X_k) = \frac{lnn}{n}$

$\therefore P(|Y_n - E(Y_n)| < \varepsilon) \geq 1 - \frac{D(Y_n)}{\varepsilon^2} = 1 - \frac{lnn}{n\varepsilon^2}$

$\therefore lim_{n\to\infty} P(|\frac{1}{n}\sum_{k=1}^{n} X_k - 0| < \varepsilon_0) = lim_{n\to\infty} 1 - \frac{lnn}{n\varepsilon^2} = 1$

$\therefore \{X_n\}$ 服从大数定律

7.

$E(\frac{1}{n}\sum_{i=1}^{n} X_i) = \frac{1}{n}\sum_{i=1}^{n} E(X_i) = \mu$

当 $|j-i| \geq 2$ 时，X_i 与 X_j 相互独立，$Cov(X_i, X_j) = 0$

$D(\frac{1}{n}\sum_{i=1}^{n} X_i) = \frac{1}{n^2} D(\sum_{i=1}^{n} X_i) = \frac{1}{n}(\sum_{i=1}^{n} D(X_i) + 2\sum_{1\leq j<k\leq n} Cov(X_i, X_j)) = \frac{1}{n^2}(\sum_{i=1}^{n} D(X_i) + 2\sum_{i=1}^{n-1} Cov(X_i, X$

$\frac{1}{n^2}(\sum_{i=1}^{n} D(X_i) + 2\sum_{i=1}^{n-1} \sqrt{D(X_i)D(X_{i+1})}) = \frac{(3n-2)\sigma^2}{n^2}$

由 Chebyshev 不等式，$\forall \varepsilon < 0$，

$1 \geq lim_{n\to\infty} P(|\frac{1}{n}\sum_{i=1}^{n} X_i - \mu| < \varepsilon) \geq lim_{n\to\infty}(1 - \frac{D(\frac{1}{n}\sum_{i=1}^{n} X_i)}{\varepsilon^2}) \leq lim_{n\to\infty}(1 - \frac{(3n-2)\sigma^2}{n^2\varepsilon^2}) = 1$

$\therefore lim_{n\to\infty} P(|\frac{1}{n}\sum_{i=1}^{n} X_i - \mu| < \varepsilon) = 1$

补充题:

(1) 因为 $X_0, X_1, ..X_n$ 是独立同分布的随机变量序列

所以设 $D(X_k) = \sigma^2, D(Y_1+Y_2+Y_3) = D(bX_0+(a+b)X_1+(a+b)X_2+aX_3) = b^2D(X_0)+(a+b)^2D(X_1)+(a+b)D(X_2)+a^2D(X_3) = (a^2+b^2+2(a+b)^2)\sigma^2$

(2)$D(\sum_{k=1}^{n} Y_k) = D(bX_0 + (a+b)\sum_{k=1}^{n-1} X_k + aX_n) = (b^2 + (n-1)(a+b)^2 + a^2)\sigma^2$

$E(\frac{1}{n}\sum_{k=1}^{n} Y_k) = \frac{1}{n}E(bX_0 + (a+b)\sum_{k=1}^{n-1} X_k + aX_n) = (a+b)\mu \therefore lim_{n\to\infty} P(|\frac{1}{n}\sum_{k=1}^{n} Y_k - (a+b)\mu| < \varepsilon) = lim_{n\to\infty}(1 - \frac{D(\frac{1}{n}\sum_{k=1}^{n} Y_k)}{\varepsilon^2}) = lim_{n\to\infty}(1 - \frac{(b^2+(n-1)(a+b)^2+a^2)\sigma^2}{n^2\varepsilon^2}) = 1$

$\therefore \{\frac{1}{n}\sum_{k=1}^{n} Y_k; n = 1, 2, 3...\}$ 依概率收敛到 $(a+b)\mu$